



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

DATA MINING

SECP2753

PROJECT 1:

Supervised Learning

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PROJECT 1: LIVER DISEASE PREDICTION

1.0 INTRODUCTION

Millions of people worldwide suffer from liver disease, a chronic and potentially fatal illness. The liver is an essential organ that produces essential proteins, detoxifies the blood and metabolizes medications and medications. When it malfunctions, the consequences can be serious, resulting in liver cancer, cirrhosis and sometimes even death. Early diagnosis is crucial in minimizing the serious consequences associated with liver illness.

In recent years, data mining and machine learning have emerged as effective tools in the healthcare industry. These strategies enable the extraction of valuable insights from massive datasets, allowing practitioners to identify high-risk individuals and make data-driven decisions. This project employs data mining techniques to evaluate a structured liver dataset with the goal of predicting liver disease based on demographic and lifestyle factors.

2.0 BUSINESS AND DATA UNDERSTANDING

2.1 PROBLEM FORMULATION

One of the most enduring issues in healthcare is diagnosing liver illness on time and accurately. Traditionally, clinical assessments, blood tests and imaging studies have been used to diagnose diseases. However, these approaches are resource-intensive and may not always be accessible. The goal of this project is to develop a predictive model that can assess whether an individual is at risk for liver disease based on existing health-related data. This process could help doctors identify high risk groups and prioritize treatments. Predictive analytics can also help to save healthcare expenses by providing preventive care rather than reactive therapies.

2.2 DATASET OVERVIEW

The liver disease dataset utilized in this project consists of 1700 individual records and 11 attributes. The dataset contains demographic information such as age and gender, lifestyle indicators such as alcohol consumption, smoking habits and physical activity as well as clinical attributes like liver function test results, BMI, and the presence of diabetes or hypertension. The target variable which is Diagnosis specifies whether liver disease exists (1) or not (0). Each attribute was chosen for its medical significance to liver health and together it provides a complete picture of an individual's risk profile.

3.0 DATA PREPARATION AND PROCESSING

It is crucial to make sure the dataset is clean, consistent, and in a readable format before developing a machine learning model. Data preparation is critical for the success of any data-driven endeavor. In this step, we prepare the liver disease dataset by running exploratory tests and deleting irrelevant or redundant entries. This procedure enhances machine learning models' interpretability, performance, and accuracy.

Steps Taken:

1. Loading and Previewing Data

```
[23]: import pandas as pd

liver_df = pd.read_csv("Liver_disease_data.csv")
liver_df.head()
```

```
[23]:
```

	Age	Gender	BMI	AlcoholConsumption	Smoking	GeneticRisk	PhysicalActivity	Diabetes	Hypertension	LiverFunctionTest	Diagn
0	58	0	35.857584	17.272828	0	1	0.658940	0	0	42.734240	
1	71	1	30.732470	2.201266	0	1	1.670557	1	0	67.309822	
2	48	0	19.971407	18.500944	0	0	9.928308	0	0	63.738956	
3	34	1	16.615417	12.632870	0	0	5.630129	0	0	64.555873	
4	62	1	16.065830	1.087815	0	1	3.566218	1	0	77.868689	

```
[25]: liver_df.info()
liver_df.describe()
liver_df.columns
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1700 entries, 0 to 1699
Data columns (total 11 columns):
#   Column              Non-Null Count  Dtype  
---  --
0   Age                  1700 non-null  int64  
1   Gender               1700 non-null  int64  
2   BMI                  1700 non-null  float64 
3   AlcoholConsumption  1700 non-null  float64 
4   Smoking              1700 non-null  int64  
5   GeneticRisk          1700 non-null  int64  
6   PhysicalActivity     1700 non-null  float64 
7   Diabetes             1700 non-null  int64  
8   Hypertension         1700 non-null  int64  
9   LiverFunctionTest    1700 non-null  float64 
10  Diagnosis            1700 non-null  int64  
dtypes: float64(4), int64(7)
memory usage: 146.2 KB
```

```
[25]: Index(['Age', 'Gender', 'BMI', 'AlcoholConsumption', 'Smoking', 'GeneticRisk',
        'PhysicalActivity', 'Diabetes', 'Hypertension', 'LiverFunctionTest',
        'Diagnosis'],
        dtype='object')
```

```
[27]: print(liver_df.head())
```

	Age	Gender	BMI	AlcoholConsumption	Smoking	GeneticRisk	PhysicalActivity	Diabetes	Hypertension	LiverFunctionTest	Diagnosis
0	58	0	35.857584	17.272828	0	1	0.658940	0	0	42.734240	1
1	71	1	30.732470	2.201266	0	1	1.670557	1	0	67.309822	1
2	48	0	19.971407	18.500944	0	0	9.928308	0	0	63.738956	0
3	34	1	16.615417	12.632870	0	0	5.630129	0	0	64.555873	1
4	62	1	16.065830	1.087815	0	1	3.566218	1	0	77.868689	1

The dataset was first loaded with `pandas.read.csv()`. Initial exploration functions such as `.head()`, `.info()` and `.describe()` were used to explore the datasets structure, data types and

statistical summary. This aided in understanding the format of the data and detecting any problems such as missing value or wrong data types.

2. Checking for Missing Values and Duplicates

```
[29]: liver_df.isnull().sum()

[29]: Age          0
      Gender        0
      BMI          0
      AlcoholConsumption 0
      Smoking       0
      GeneticRisk   0
      PhysicalActivity 0
      Diabetes      0
      Hypertension  0
      LiverFunctionTest 0
      Diagnosis     0
      dtype: int64

[31]: liver_df.duplicated().sum()
      liver_df.drop_duplicates(inplace=True)
```

The dataset was checked for missing values with `.isnull().sum()`, revealing the presence of null entries. Duplicate entries were found with `.duplicated().sum()` and eliminated with `.drop_duplicates()` to guarantee that each data point was unique, enhancing data quality and reducing redundancy.

3. Data Type Conversion

```
[33]: liver_df.dtypes

# Convert to categorical where appropriate
liver_df['Gender'] = liver_df['Gender'].astype('category')
liver_df['Diagnosis'] = liver_df['Diagnosis'].astype('category')
```

The Gender and Diagnosis columns have been transformed from object/string to categorical types. This conversion is advantageous because it saves memory and enables machine learning models to correctly comprehend categorical variables during training.

4. Visualizing Outliers with a Boxplot



A boxplot was built to show the distribution and potential outliers in numerical data including BMI, alcohol intake, and liver function test results. This stage assisted in finding extreme values that could impact model performance or indicate data issues.

5. Feature Scaling

```
[37]: from sklearn.preprocessing import StandardScaler

liver_scaled = pd.DataFrame(
    StandardScaler().fit_transform(liver_df.select_dtypes(include='number')),
    columns=liver_df.select_dtypes(include='number').columns
)
```

Feature scaling is important because it ensures that all numerical features are on the same scale, which is especially beneficial for distance-based models such as KNN or logistic regression.

6. Exporting the Cleaned Dataset

```
[39]: liver_df.to_csv("cleaned_liver_dataset.csv", index=False)
```

Finally, the cleaned and preprocessed dataset was saved as a new CSV file for later use in model training or analysis. This ensures repeatability and allows us to work with consistent data across development stages.

4.0 MODEL DEVELOPMENT

4.1 OVERVIEW

In this dataset, we used Logistic Regression, a supervised learning algorithm. This method is a statistical-based classifier that is ideal for binary classification tasks such as predicting whether a person has liver disease (diagnosis = 1) or not (diagnosis = 0). The holdout method, which divides the dataset into two parts which are training and testing data that were used in this model development process. This method allows us to train the model on one segment of the data and then evaluate it on a different, unseen segment to check its accuracy.

4.2 LOGISTIC REGRESSION METHOD

The steps are as following:

1. Import necessary libraries:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score,
recall_score, f1_score
```

2. Load the cleaned dataset

```
df = pd.read_csv("cleaned_liver_dataset.csv")
```

3. Separate features (X) and target label (y)

```
X = df.drop("Diagnosis", axis=1)
y = df["Diagnosis"]
```

4. Split the data using holdout method: testing - 20% , training - 80%

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,  
random_state=42)
```

5. Scale the features: standardized numerical features such as Age, BMI, AlcoholConsumption, PhysicalActivity and LiverFuntionTest

```
scaler = StandardScaler()  
X_train_scaled = scaler.fit_transform(X_train)  
X_test_scaled = scaler.transform(X_test)
```

6. Train the Model: used LogisticRegression() from the sklearn.linear_model library
from sklearn.linear_model import LogisticRegression

```
model = LogisticRegression()  
model.fit(X_train_scaled, y_train)
```

7. Make predictions: the trained model was used to predict class labels on the test data

```
y_pred = model.predict(X_test_scaled)
```

4.3 RESULT

1. Model Used: Logistic Regression
2. Training size: 1360 records
3. Testing size: 340 records

5.0 MODEL EVALUATION

5.1 EVALUATION METRICS

Confusion matrix:

```
conf_matrix = confusion_matrix(y_test, y_pred)
```

= Confusion Matrix:

```
[[124(TN) 38(FP)]
```

```
 [ 26(FN) 152(TP)]]
```

Based on the Confusion matrix:

Metric	Result
Accuracy [(TP + TN) / Total]	$\frac{(152 + 124)}{340} = 81.2\%$
Precision [TP / (TP + FP)]	$\frac{152}{(152 + 38)} = 80\%$
Recall [TP / (TP + FN)]	$\frac{152}{(152 + 26)} = 85.4\%$
F1 Score [2*(Precision*Recall) / (Precision + Recall)]	$2 \times \left(\frac{0.8117 \times 0.8539}{(0.8117 + 0.8539)}\right) = 82.6\%$

5.2 PERFORMANCE ANALYSIS

- High recall (85%): in a medical setting, the model's ability to identify the majority of liver disease cases is crucial and effective
- Good Precision (80%): liver disease predictions are reasonably accurate
- F1 Score (82.6%): accomplished recall and precision in equal measure
- The model's overall accuracy of 81.2% indicates that it generalizes well

5.3 RESULT REPORT

When it comes to liver disease prediction, the logistic regression model performs exceptionally well particularly when it comes to detecting true positive instances. It scores higher than 80% on every important evaluation metric. This indicates its efficiency for early liver disease prediction based on lifestyle and demographic variables.